

MATTHEW NUTT

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EDUCATION

Rice University - Master's in Electrical & Computer Engineering

Expected Graduation May 2027

August 2022 - Present

Houston, TX

- ECE Future Star Scholarship
- GPA: 3.89 / 4.00

EXPERIENCE

Electrical Engineering Intern

Lutron Electronics

May - August 2025

Austin, TX

- Designed circuit schematics and PCB layouts optimized for low-power solar energy harvesting.
- Achieved 20% solar conversion efficiency in ideal conditions, generating $15\times$ more power than required.

Chief Engineer

Rice Robotics Club

April 2024 - April 2025

Houston, TX

- Contributed to the management of a lunar rover project, cat robot project, and BattleBots team, encompassing about 100 individuals from various engineering disciplines.
- Corresponded with team leads on project timelines, technical decisions, and specific team needs.

PROJECTS

FPGA Linear System Solver

December 2025

- Implemented a CORDIC-based linear solver algorithm for 4×4 matrices in C++, synthesized the system via Vitis HLS and AMD Vivado, and tested the design on an AUP-ZU3 FPGA development board.
- Achieved $29.5\mu\text{s}$ latency with an average deviation of 0.097% from expected results.

Snake Game ASIC

May 2025

- Programmed a basic snake game in Verilog, featuring a multiplexed 8×8 display grid, randomly generated "food" positions, and dual-phase clocking.
- Verified each module's functionality with dedicated simulation testbenches on QuestaSim.
- Generated layout files for the AMI $0.5\mu\text{m}$ PDK from synthesized logic and manually routed I/O to a pre-made pad frame using Magic to prepare for tapeout.

32-bit Carry-Bypass Adder

April 2025

- Constructed a schematic-level design of a 32-bit carry-bypass adder for the Skywater 130 nm PDK in Cadence Virtuoso, with eight bypass units and transistor/gate sizing optimization for each unit.
- Achieved 640 ps critical path delay and $133\mu\text{W}$ average power consumption.

Power Supply Supervisor

December 2024

- Created a schematic-level ASIC of a power supply supervisor that monitors five voltage rails and generates a synchronous powergood output.
- Designed a 500 kHz PLL circuit in LTSpice that achieved lock in 0.66 ms and produced 0.155% clock jitter.

SKILLS

Programming Languages

C/C++, Verilog, Python, MATLAB, Java

Design Tools

Cadence Virtuoso and Innovus, Vitis HLS, AMD Vivado, Siemens HyperLynx, Keysight ADS, Magic, KiCad, LTSpice